Property:

If $X_1, X_2, ..., X_n$ are independent r.vs, X_i having exponential distribution with parameters $\alpha_i, i = 1,2,3...,n$ then $Z = \min(X_1, X_2, ..., X_n)$ has exponential distribution with parameter $\sum_{i=1}^{n} \alpha_i$

mgf of Exponential distribution

$$M_X(t) = \int_0^\infty e^{tx} f(x) dx = \int_0^\infty e^{tx} \alpha e^{-\alpha x} dx = \frac{\alpha}{\alpha - t}, \alpha > \text{t}$$

Hence Mean =

$E(X) = M_x'(t)$ at $t = 0$ simillary varianceis $\sigma^2 = E(X^2) - \{E)X)\}^2 = M_x''(t) - \{M_x'(t)\}^2$ at $t = 0$